

Questions

[SET A]

Design the **Pirate** class as a subclass of **Character** so that it works with the given **Tester** class and produces the correct output.

Note: When Haki goes above 1500 the Pirate's rank becomes Pirate Emperor.

Anytime power is used, health decreases by 20 as well as the haki spent.

Driver Code	Output
<pre>public class Tester { public static void main(String[] args) { Pirate p = new Pirate("Luffy", 200, 1300); System.out.println(p.info()); System.out.println("1====="); p.usePower(20); System.out.println("2====="); System.out.println(p.info()); System.out.println("3====="); p.restoreHaki(250); System.out.println("4====="); System.out.println(p.info()); System.out.println("5====="); p.usePower(50); System.out.println("6====="); System.out.println(p.info()); } }</pre>	<pre>Pirate Character: Luffy Health: 200 haki: 1300 Rank: Captain 1===== Luffy attacks with 20 haki! 2===== Pirate Character: Luffy Health: 180 haki: 1280 Rank: Captain 3===== Luffy regains 250 haki. Luffy has ascended to Pirate Emperor rank! 4===== Pirate Character: Luffy Health: 180 haki: 1530 Rank: Pirate Emperor 5===== Luffy attacks with 50 haki! 6===== Pirate Character: Luffy Health: 160 haki: 1480 Rank: Pirate Emperor</pre>
<pre>public class Character { private String name; public int health; public Character(String name, int health) { this.name = name; this.health = health; } public String getName() { return name; } public String info() { return "Character: " + name + "\nHealth: " + health; } }</pre>	

[SET B]

Design the **Ninja** class as a subclass of **Character** so that it works with the given **Tester** class and produces the correct output.

Note: When Chakra goes above 1100 the Ninja's rank becomes Chunin.

Anytime a jutsu is used, health decreases by 15 as well as the chakra spent.

Driver Code	Output
<pre>public class Tester { public static void main(String[] args) { Ninja n = new Ninja("Naruto", 90, 1000); System.out.println(n.info()); System.out.println("1====="); n.useJutsu(20); System.out.println("2====="); System.out.println(n.info()); System.out.println("3====="); n.boostChakra(200); System.out.println("4====="); System.out.println(n.info()); System.out.println("5====="); n.useJutsu(50); System.out.println("6====="); System.out.println(n.info()); } }</pre>	<pre>Ninja Character: Naruto Health: 90 chakra: 1000 Rank: Genin 1===== Naruto uses a jutsu with 20 chakra! 2===== Ninja Character: Naruto Health: 75 chakra: 980 Rank: Genin 3===== Naruto regains 200 chakra. Naruto has ascended to Chunin rank! 4===== Ninja Character: Naruto Health: 75 chakra: 1180 Rank: Chunin 5===== Naruto uses a jutsu with 50 chakra! 6===== Ninja Character: Naruto Health: 60 chakra: 1130 Rank: Chunin</pre>
<pre>public class Character { private String name; public int health; public Character(String name, int health) { this.name = name; this.health = health; } public String getName() { return name; } public String info() { return "Character: " + name + "\nHealth: " + health; } }</pre>	

[Solutions - A]

```
public class Pirate extends Character { // 0.5
    public int haki; // 0.5
    public String rank; // 0.5

    // 2
    public Pirate(String name, int health, int haki) {
        super(name, health);
        this.haki = haki;
        this.rank = "Captain";
    }

    // 2
    public String info() {
        return "Pirate " + super.info()
            + "\nhaki: " + haki
            + "\nRank: " + rank;
    }

    // 1.5
    public void usePower(int amount) {
        System.out.println(getName() + " attacks with " + amount + " haki!");
        haki -= amount;
        health -= 20;
    }

    // 3
    public void restoreHaki(int amount) {
        haki += amount;
        System.out.println(getName() + " regains " + amount + " haki.");

        if (haki > 1500) {
            rank = "Pirate Emperor";
            System.out.println(getName() + " has ascended to Pirate Emperor rank!");
        }
    }
}
```

[Solutions - B]

```
public class Ninja extends Character { // 0.5
    public int chakra;                // 0.5
    public String rank;                // 0.5

    // 2
    public Ninja(String name, int health, int chakra) {
        super(name, health);
        this.chakra = chakra;
        this.rank = "Genin";
    }

    // 2
    public String info() {
        return "Ninja " + super.info()
            + "\nchakra: " + chakra
            + "\nRank: " + rank;
    }

    // 1.5
    public void useJutsu(int amount) {
        System.out.println(getName() + " uses a jutsu with " + amount + " chakra!");
        chakra -= amount;
        health -= 15;
    }

    //3
    public void boostChakra(int amount) {
        chakra += amount;
        System.out.println(getName() + " regains " + amount + " chakra.");

        if (chakra > 1100) {
            rank = "Chunin";
            System.out.println(getName() + " has ascended to Chunin rank!");
        }
    }
}
```

[Rubric]

Criteria	Points
Class Design & Syntax → class name and use of extends	0.5
Instance Variables → 2 instance variables	1
Constructor → [1] Use of super() with correct parameters → [1] Update instance variables	2
info(): String → [0.5] Use of super.info() → [0.5] Concatenation of instance variables → [0.5] Correct method signature → [0.5] Correct return type	2
usePower(int amount) / useJutsu(int amount): void → [0.5] Use of getName() in print → [0.5] Deduction of haki / chakra by amount → [0.5] Deduction of health	1.5
restoreHaki(int amount) / boostChakra(int amount): void → [1] Proper print statements with getName() → [0.5] Increase haki / chakra by amount → [1] If haki / chakra reaches above cutoff value change rank → [0.5] Print updated rank	3
Total	10